

iDEC Responsible Research Form

Dear iDEC Teams,

This is the iDEC 2026 Responsible Research Form. Submission of this form is mandatory for your participation in iDEC. Please remember: safety and ethical considerations are an important part of modern research and we are not asking you these questions to waste your time, but to educate you, and protect you and those around you! Please take the time to answer the questions as a team, and if you need help, do not hesitate to contact the Commission for Responsible Research (vinke@idechq.org). We are here to help you with things like this!

The Responsible Research Form should be submitted prior to the start of the wetlab, but no later than July 1. If you find that you need to change parts of your projects (e.g., use additional genes), please contact us and we will approve those changes as soon as possible.

The responsible research form consists of 5 sections: General Information about your team, Laboratory Biosafety, Genes and Chassis, Biosecurity, and Ethical Considerations. We have gathered resources for each section to guide your team, and we have completed the entire form as an example on our website (https://idec.io/pages/ethics_biosafety_and_biosecurity.html).

Be aware that you are able to shape the values of your research community. In our ideal directed evolution research community, safety, security, and ethical considerations are valued and common place. We hope the iDEC responsible research project can teach you how important these values are. We hope you learn more about your project as you think about biosafety, biosecurity, and ethical considerations, and have a wonderful time during iDEC 2026!

Your Commission for Responsible Research

Team Name *

OUC-DE

Email address for questions regarding your Responsible Research Form *

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What is the research hypothesis of your project? *

We hypothesize that by engineering and coupling MddH (S-methyltransferase) and Tmm (trimethylamine monooxygenase), we can establish a highly efficient synthetic metabolic pathway for the high-value valorization of toxic industrial hydrogen sulfide (H₂S). Specifically, we hypothesize that targeted mutagenesis and directed evolution will overcome natural enzymatic bottlenecks, enabling the continuous, high-yield upcycling of an environmental pollutant directly into dimethyl sulfoxide (DMSO), a premium industrial solvent.

Please provide a detailed abstract of your project. *

Industrial hydrogen sulfide (H₂S) emissions cause severe ecological hazards and waste abundant sulfur resources. Traditional industrial technologies only eliminate H₂S pollutants without resource recycling, making the high-value utilization of waste H₂S an urgent demand. This study aims to overcome the catalytic defects of natural enzymes to realize efficient biological recycling of H₂S. A continuous biocatalytic cascade system is constructed using engineered MddH and optimized Tmm. The modified MddH catalyzes sulfide methylation, and the optimized Tmm further oxidizes the generated intermediates, enabling the efficient conversion of toxic HS into dimethyl sulfoxide (DMSO), a high-value industrial solvent. Different from conventional chassis cell modification strategies, this work focuses on optimizing core catalytic steps. Combining site-directed mutagenesis and directed evolution, we improve the active-site microenvironment and substrate affinity of the two enzymes. This study establishes an efficient biomanufacturing pathway for waste gas valorization. By transforming the traditional pollution control model into resource reuse, this work provides an innovative, economically feasible synthetic biology strategy for industrial HS treatment and high-value utilization.

State-of-knowledge

The following four questions will ask you how much you knew about risk assessment before iDEC. You don't have to answer them, but they would help our committee design better, more tailored educational projects. There are no wrong answers to these questions and it is okay if you answer "no" to all of them - it is not your fault if no one told you about these risks.

Have you ever assessed biosafety risks of a scientific project? When was the first time and place someone taught you about biosafety risks?

Yes, we have assessed the biosafety risks of our project. I first learned about biosafety risks in May 2026 during the mandatory Laboratory Safety Training Course at Laboratory for Microbial Oceanography

Have you ever assessed biosecurity risks of a scientific project? When was the first time and place someone taught you about the difference between biosafety and biosecurity risks?

Yes. Biosafety focuses on protecting humans and the environment from harmful biological agents (containment), while Biosecurity focuses on protecting biological materials from loss, theft, misuse, or intentional release (control and asset protection). I first learned this difference in few years ago during a Laboratory Safety Training.

Do you know what "Dual-Use Research of Concern " means? Where and when did you learn about dual-use research of concern?

Yes, the dual-use refers to life sciences research that, based on current understanding, can be reasonably anticipated to provide knowledge, information, products, or technologies that could be directly misapplied to pose a significant threat to public health, safety, or the environment. We learned about this concept through the iDEC safety guidelines and our institutional laboratory biosafety training.

Do you know what "Gain of function experiment" means? Where and when did you learn about them?

Yes, GoF experiments involve enhancing the prevalence, transmissibility, or virulence of potential pandemic pathogens. I learned about this concept in 2026 during laboratory biosafety training and molecular biology class. Our project modifies non-pathogenic laboratory strains for industrial/ecological purposes and does not enhance any hazardous traits.

Laboratory Biosafety

Thank you for helping us by sharing your level of knowledge prior to iDEC! We will now start with the risks you have identified for your iDEC project. The first category of risks is "Laboratory Biosafety." In this section, you will need to indicate that your lab meets the criteria to be a safe place for you to work, and ensure no accidental release of your GMOs.

Biosafety describes personal protection and protection from accidental release of biological agents that have the potential to harm plants, animals, humans, or the environment. In this laboratory biosafety section of the Responsible Research Form, your teams will need to answer questions about safety equipment in your lab and safety countermeasures to prevent accidental release of GMOs.

If you are looking for resources or would like to learn more, we recommend you take a look at the WHO Laboratory Biosafety Manual:

<https://www.who.int/publications/i/item/9789240011311>

Mention the PI of your team and the relevant qualifications of your PI to function as your team's person responsible for safety. *

Dr. Yunhui Zhang is an Associated Professor at Ocean University of China and serves as the Principal Investigator of our team. She has extensive experience in microbiology, molecular biology, and synthetic biology research. She has led four research projects, including those funded by the National Natural Science Foundation of China, and has supervised eight graduate researchers.

As the person responsible for laboratory safety, Prof. Zhang has many years of experience in laboratory management and biosafety supervision. She ensures that all experiments comply with institutional biosafety regulations and relevant national guidelines. All team members receive appropriate laboratory safety training before conducting experiments.

Did you get permission from your institute to perform the experiments necessary for your iDEC project? *

Yes, we have obtained official permission and safety clearance from the Laboratory for Microbial Oceanography to conduct all experiments related to our iDEC project.

List all topics mentioned in your safety introduction. *

Laboratory standard operating procedures (SOPs)
Personal Protective Equipment (PPE) usage (lab coats, gloves, goggles)
Biological waste disposal and autoclaving protocols
Emergency response (chemical spills, fire, accidental exposure)
Biosafety Level 1/2 material handling rules

What is the biosafety level of your laboratory? *

☐ We have a computational project and don't use a laboratory

☒ BSL1

☐ BSL2

☐ BSL3

☐ Other:

Describe the safety measures in your laboratory (e.g. eye shower, emergency shower, etc) *

Our laboratory is equipped with standard safety facilities including an emergency eye shower, a full-body emergency shower, autoclaves for biohazard waste treatment, fire extinguishers, first-aid kits, and clear signs for emergency exits. All personnel are required to wear appropriate PPE (lab coats and gloves) at all times.

Describe how GMOs are prevented from being unintentionally released into the environment, including waste management *

To strictly prevent the unintentional release of genetically modified organisms (GMOs) into the environment, our team strictly adheres to standard laboratory biosafety protocols through the following measures:

·Physical Containment & Standard Practices: All experimental procedures involving engineered strains are confined strictly within designated biosafety cabinets or laminar flow hoods. Standard Personal Protective Equipment (PPE) including lab coats, disposable gloves, and safety goggles are mandatory and must not be worn outside the laboratory area. Windows are kept closed, and access to the lab is restricted to authorized personnel only.

·Liquid Waste Management: All liquid cultures and waste containing engineered bacteria/yeast are accumulated in dedicated, labeled biohazard containers. Before disposal, they undergo high-pressure steam sterilization (autoclaving at 121 °C for at least 20 – 30 minutes) or are chemically inactivated using a freshly prepared sodium hypochlorite solution (bleach) with sufficient contact time to ensure complete cell lysis and DNA degradation.

·Solid Waste Management: Solid biological waste, including contaminated pipette tips, culture plates, microcentrifuge tubes, and gloves, is collected in leak-proof, color-coded biohazard bags. All solid waste is autoclaved at 121 °C to ensure absolute sterilization before entering the institutional municipal waste stream.

·Emergency Spill Protocol: In the event of an accidental spill, the affected area will be isolated immediately. We utilize a standard spill kit containing absorbent materials and disinfectants (such as 75% ethanol or chlorine-containing disinfectants) to neutralize and clean the spill, and the cleanup waste is treated as hazardous biological waste.

Do you work with animals or samples (urine, blood, saliva, etc) derived from animals or humans? *

No

Are you working with chemicals that are mutagenic, carcinogenic, explosives, or narcotics?

If yes indicate what chemicals, in what amounts you use them, and what safety measures are in place. *

We use ethidium bromide to stain our DNA gels and acrylamide to prepare our SDS-PAGE protein gels. We have a special designated area with additional safety precautions (including fume hood operation, double nitrile gloves, and dedicated waste containers) to prevent accidental exposure of the researchers working in the lab to these chemicals. We only keep small stocks in the lab: a 5 mL stock of ethidium bromide and less than 100 mL of acrylamide stock solutions.

Genes and Chassis

Genes and chassis of each team must be biosafety level 1 or 2. Any genes and chassis from organisms that fall into the biosafety level 3 or 4 category are prohibited to be used in iDEC. There will be a whitelist of chassis and all genes and chassis used beyond that must be listed here.

Whitelists are kind of outdated when it comes to synthetic biology, since the taxonomic classification of organisms into safety groups does not necessarily reflect the risks of an organism to cause harm (e.g., an organism might carry genes for toxins that would not be expressed under the conditions used to characterize that organism, so the recommended biosafety level would be low, but we can still derive the toxin from that organism and express it recombinantly). That said, we decided to provide a short whitelist for iDEC so you don't have to list the usual lab workhorses like your

Escherichia coli lab strain. Our whitelist includes the FDA's GRAS organism as well as commonly used organisms for directed evolution projects. The iDEC whitelist can be found in this form and on the iDEC website and any organisms, genes from this white list do not need to be listed here unless you are evolving them

(https://idec.io/pages/ethics_biosafety_and_biosecurity.html).

For us, it is very important that you really think about what additional risks might come from your evolved molecule (or pathway, or genome). The thing about directed evolution is that you are creating something novel, and that novel product may not have the same level of safety as the scaffold you started with.

List all chassis you plan to use that are not listed in the iDEC whitelist below, as well as their biosafety level (with reference) and a short explanation how you plan to use these. *

<i>Acetobacter suboxydans</i>	<i>Eisenia bicyclis</i>	<i>Lactococcus lactis</i>	<i>Porphyra suborbiculata</i>
<i>Acetobacter xylinum</i>	<i>Endothia parasitica</i>	<i>Laminaria angustata</i>	<i>Porphyra crispata</i>
<i>Actinoplanes missouriensis</i>	<i>Eremothecium ashbyii</i>	<i>Laminaria claustraria</i>	<i>Porphyra tenera</i>
<i>Anolis japonicus</i>	<i>Escherichia coli BL21 (DE3)</i>	<i>Laminaria digitata</i>	<i>Rhizopus niveus</i>
<i>Aspergillus niger</i>	<i>Escherichia coli DH5alpha</i>	<i>Laminaria japonica</i>	<i>Rhizopus oryzae</i>
<i>Aspergillus niger</i> var. <i>Awamori</i>	<i>Escherichia coli DH5alpha F'</i>	<i>Laminaria longiruris</i>	<i>Rhodomenis palmata</i>
<i>Aspergillus oryzae</i>	<i>Escherichia coli TOP10</i>	<i>Laminaria longissima</i>	<i>Saccharomyces cerevisiae</i>
<i>Arabidopsis thaliana WT</i>	<i>Eucheuma cottonii</i>	<i>Laminaria ochotensis</i>	<i>Saccharomyces fragilis</i>
<i>Cooney et Emerson</i>	<i>Eucheuma spinosum</i>	<i>Laminaria saccharina</i>	<i>Scytosiphon lome</i>
<i>Bacillus cereus</i>	<i>Furcellaria fastigiata</i> of the class <i>Rhodophyceae</i> (red seaweed)	<i>Leuconostoc citovorum</i>	<i>Streptococcus cremoris</i>
<i>Bacillus coagulans</i>	<i>Fusarium moniliforme</i>	<i>Leuconostoc dextranicum</i>	<i>Streptococcus lactis</i> subspecies <i>diacetylactis</i>
<i>Bacillus licheniformis</i>	<i>Gigartina acicularis</i>	<i>Leuconostoc mesenteroides</i> strain NRRL B-512(F)	<i>Streptococcus lactis</i>
<i>Bacillus stearothermophilus</i>	<i>Gigartina pistillata</i>	<i>Macrocystis pyrifera</i>	<i>Streptococcus thermophilus</i>
<i>Bacillus subtilis</i>	<i>Gigartina radula</i>	<i>Mortierella vinaceae</i> var.	<i>Streptomyces chattanoogensis</i>
brown algae	<i>Gigartina stellata</i>	<i>Raffinoseutilizer</i>	<i>Streptomyces griseus</i>
<i>Candida guilliermondii</i>	<i>Gloiopeltis furcata</i>	<i>Mucor miehei</i>	<i>Streptomyces natalensis</i>
<i>Candida lipolytica</i>	<i>Hizikia fusiforme</i>	<i>Mucor miehei</i> var. <i>Cooney et Emerson</i>	<i>Streptomyces olivaceus</i>
<i>Candida pseudotropicalis</i>	<i>Kjellmaniella gyrate</i>	<i>Mucor pusillus</i> Lindt	<i>Streptomyces olivochromogenes</i>
<i>Candida utilis</i>	<i>Kluyveromyces lactis</i>	<i>Penicillium raquefortii</i>	<i>Streptomyces rubiginosus</i>
<i>Chlamydomonas reinhardtii</i>	<i>Kluyveromyces marxianus</i> var. <i>lactis</i>	<i>Petalonia fascia</i>	<i>Xanthomonas campestris</i>
<i>Chondrus crispus</i>	<i>Lactobacillus bulgaricus</i>	<i>Porphyra deusta</i>	
<i>Chondrus ocellatus</i>	<i>Lactobacillus fermentum</i>	<i>Porphyra perforata</i>	

E. coli ER2738 (BSL-1, reference: New England Biolabs strain genotype)

List all genes and plasmids you plan to use that are not derived from one of the organism on the iDEC whitelist above, as well as their original host/lab and their biosafety level of their original host (with references) and a short explanation on how you plan to use these. *

pET28a(+) plasmid from Escherichia coli BSL-1 (Novagen/Merck, GenBank MF973384.1). Used as expression vector for MddH mutants. It carries a T7 promoter and kanamycin resistance gene. The plasmid will be transformed into E. coli BL21(DE3) for IPTG-inducible expression of MddH variants.

pCDFDuet-1 plasmid from Escherichia coli BSL-1 (Novagen/Merck, GenBank DQ440487.1). Used as co-expression vector for Acul and Tmm. It carries dual T7 promoters and chloramphenicol resistance. The plasmid will be co-transformed with pET28a-MddH for three-enzyme co-expression.

p15A-trc-Scaffold plasmid from Escherichia coli BSL-1 (p15A replicon, GenBank L08813.1). Used to express GBD-SH3-PDZ scaffold protein for co-localization of MddH, Tmm and Acul to achieve metabolic channeling.

acul gene from Ruegeria pomeroyi BSL-1 (ATCC 700808, <https://bacdive.dsmz.de/strain/13387>). Encodes Acul enzyme that converts acrylate to propionyl-CoA for detoxification. The gene will be cloned into pCDFDuet-1 and co-expressed with Tmm.

Which genes are evolved in your experiment? What kind of improvement do you want to achieve by your evolution (e.g. increase activity, increase stability)? *

We evolve mddH to increase catalytic activity and substrate affinity toward HS/MeSH for efficient methylation, and tmm to enhance substrate specificity toward DMS for efficient DMSDMSO oxidation. The overall goal is to couple both enzymes into a continuous cascade that converts toxic industrial HS waste into high-value DMSO.

Biosecurity

Biosecurity describes the prevention of biological agents from unauthorized access, theft, misuse or loss. In the biosecurity section of the responsible research form, your team is required to evaluate potential biosecurity risks and

their countermeasures. In this section we will ask for the dual-use research of concern potential. Dual-use research of concern in the life sciences, describes the potential of your project, generated technologies and knowledge to be directly misused by someone else to cause serious harm to animals, humans, plants, national security or the environment. Please take this section seriously. We have seen a lot of teams becoming very creative when it comes to envision 10,000 benefits of their project, but we almost never become creative when it comes to envision the risks. That is something we need to change! For most risks you identify, you will notice that countermeasures can be easily integrated and that makes your project better and more likely to be used in a project beyond your lab bench!

If you would like to learn more, take a look at the WHO biorisk management guidance

(https://www.who.int/ihr/publications/WHO_CDS_EPR_2006_6/en/), the NIH dual-use research of concern information page (<https://osp.od.nih.gov/biotechnology/dual-use-research-of-concern/>) or attend the free FutureLearn "Next Generation Biosecurity: Responding to 21st Century Biorisks" course (<https://www.futurelearn.com/courses/biosecurity>).

Describe the access control measures of your laboratory. *

Access to our laboratory is restricted to authorized personnel, including faculty members, laboratory staff, and trained students. The laboratory is located in a controlled-access building, and entry is permitted only through an electronic access control system (campus card/key access).

Undergraduate team members are allowed to work in the laboratory only after completing laboratory safety training and under the supervision of authorized instructors or graduate students.

Biological materials, chemicals, and laboratory equipment are stored in designated areas with restricted access. Hazardous reagents and biological samples are securely stored and managed according to institutional biosafety regulations. Laboratory activities comply with the safety policies of Ocean University of China, and all personnel are required to follow standard operating procedures (SOPs) and wear appropriate personal protective equipment (PPE).

Evaluate if your project has Dual Use Research of Concern potential. *

Enzyme engineering and directed evolution technologies have general DURC potential, as they could theoretically be misused to modify harmful biological molecules. However, our specific application — converting industrial HS waste into DMSO for environmental remediation — has clear beneficial use and involves only non-pathogenic enzymes acting on small sulfur compounds, not on any pathogens or toxins. Given that all techniques used are standard and widely available, we categorize our project as low DURC potential. We note that theoretical misuse risk exists for any protein engineering work, but our experiments do not pose additional risk beyond existing public knowledge.

If yes, indicate what measures your team takes to minimize the risk of misuse. *

We do not consider our project to have high DURC potential. Nevertheless, we take precautions to minimize any theoretical risk of misuse by emphasizing our environmental remediation purpose in all communications, using only non-pathogenic BSL-1 materials, requiring all team members to complete safety training, and conducting all experiments in designated access-controlled laboratory areas with proper storage of materials. Our PI and team will maintain continuous dual-use awareness throughout the project.

Ethical Considerations

Identifying ethical issues of scientific research projects is a big part of responsible research. To be able to identify possible unethical practices, teams need to evaluate who is impacted by their project and what this influence means to

the person, group or society as a whole.

In case you want to do research on human subjects (this includes for example interviews or surveys) we have included a questionnaire to identify possible issues that you need to evaluate if your experimental questions validates the ethical concern of pursuing this experiment. This questionnaire should give you an idea about the broad range of considerations necessary when you want to conduct research on human subjects. We do still encourage you to always get approval by your institute's ethic council if you would like to conduct such experiments.

If you would like to learn more about bioethics, we recommend to take a look at the NIH or UNESCO bioethic websites (NIH: <https://www.niehs.nih.gov/research/resources/bioethics/index.cfm> UNESCO: <https://en.unesco.org/themes/ethics-science-and-technology/bioethics>).

List all people/groups that are potentially influenced by your research or outcomes of your research and describe the impact of your project on these people/groups. *

Our project focuses on establishing a biocatalytic cascade system for converting HS to DMSO through enzyme engineering, which is more of a proof-of-concept for technical feasibility rather than an immediate industrial application. Therefore, our research findings will mainly influence researchers in synthetic biology and environmental biotechnology — we hope our work will provide a valuable reference for industrial waste gas valorization, and may also inspire research teams working on sulfur metabolism or enzyme engineering. If our technology is considered for actual industrial deployment in the future, we will re-evaluate the associated environmental release risks, biosafety concerns, and ethical issues prior to implementation. In addition, our own team members, as the direct executors of the experiments, will also be affected — we need to undergo safety training, follow standard operating procedures, and enhance our research capabilities throughout the project. At present, we plan to keep our research confined to laboratory-scale fundamental exploration.

Do you see any ethical issues arising from your project? If so, indicate why. *

We do not see any ethical issues arising from our project.

Do you plan to conduct research on humans or human samples (this includes questionnaires, interviews, public engagement)? *

No, we do not plan to conduct any research on humans or human samples, including questionnaires, interviews, or public engagement.

If yes, does your experiment fulfil all requirements to be categorized as research with low potential to cause ethical issues? (A good guideline is the questionnaire below. If you can answer all questions on the questionnaire below with no, your research probably has a low potential to be the cause of ethical issues.) *

This questionnaire is derived from the basic questionnaire of the ethics council of Bielefeld University.

Please answer all 12 questions by ticking yes or no as appropriate:	DGPs*	yes	no
1. Will members of a vulnerable group or people who cannot give their own consent participate in this study (e.g., children and adolescents under 16 years of age, people with a learning disability, people in psychotherapeutic treatment)?	3 (b)		
2. Will it be necessary that people participate in this study without having been informed about this previously or without having given their consent to participate (e.g., as in covert observation)?	6		
3. Will the study involve covert observation or any other method that precludes informed consent, full debriefing, or the opportunity for participants to have their data deleted?	3, 9		
4. Will the study feature questions about topics that are of an intimate nature or that participants may perceive as stigmatizing (e.g., questions pertaining to illegal or deviant behavior or to sexual preferences)?	3 (d)		
5. Does the study include an active deception of participants or will information be deliberately withheld from participants? (This does not apply to withholding the study hypothesis.)	8		
6. Is there a risk that the study may cause psychological stress, fear, exhaustion, or other negative effects in participants to an extent that they would not normally encounter in their daily life?	3 (d), 9		
7. Is there a risk that the study may cause participants to experience pain or more than mild discomfort?	3 (d), 9		
8. Will the participants be given drugs, placebos, or other substances to ingest (e.g., food, beverages, vitamins), or will the participants be subjected to any invasive or potentially harmful procedures?	3 (d), 3 (e), 8, 9		
9. Will video or audio recordings be taken without prior consent by the participants ?	3, 4		
10. Will any bodily substances of participants be sampled (blood, saliva, etc.)?	3, 4		
11. Will participants receive a payment of more than 10 Euros <i>per hour</i> for their participation?	7		
12. Is there a conflict of interest for the applicant or applicants because of (a) economic or personal connections to a contracting entity or a collaborator whose interests may be affected positively or negatively by results of the research, or (b) any other factor(s) that might affect the applicants' independent scientific judgment?	–		

Note: * This column points out particularly relevant subsections of section 7.3 of the document "Berufsethische Richtlinien der DGPs und des BDP". See website of the EUB.

we do not plan to conduct any research on humans or human samples.

Thank you!

We hope this responsible research form gave you some idea how complex safety, security and ethical considerations for a research project are and that you learned a little bit more about your project by filling out this form. Two last questions:

We would like to ask you about the time your team needed to fill out the form and if you would consider the effort as too much.

Would it be okay if we use your answers to learn more about risk assessment of directed evolution projects? We might give a talk at the festival where we would like to talk about case studies or publish about the iDEC risk assessment resources. This does not include your email address or PI information and we would of course anonymize all answers, but someone might be able to connect projects to answers given in the form.

How much time did you and your team need to consider risks and to fill out the form. Would you consider this time as too long, too short or adequate?

Adequate. Our team spent approximately 4 to 5 hours in total on this process, including about 2 to 3 hours for group discussion of potential risks and another 1 to 2 hours for filling out the form and cross-checking the required information.

Do you give consent for your answers in this form to be anonymously used for talks and publications by the iDEC Commission for Responsible Research? *

☒ Yes

☐ No

This form was created inside iDEC.

Google Forms